

Car Sales Dashboard

Case Study Report

An Interactive Excel Dashboard Built with Pivot Tables, Pivot Charts, KPI Cards and Slicers

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Tool: Microsoft Excel
Dataset Source: Kaggle

Project Overview

The Car Sales Dashboard is an interactive dashboard developed using Microsoft Excel to analyze and visualize car-related data. The project utilizes the Cars Dataset from Kaggle and transforms raw data into meaningful insights using Pivot Tables, Pivot Charts, KPI Cards, and Slicers.

The dashboard enables users to compare different car brands, analyze pricing trends, study fuel type and transmission distributions, and evaluate vehicle specifications through interactive visualizations.



Final Car Sales Dashboard

Problem Statement

Analyzing a large amount of vehicle data manually is difficult and time-consuming. Raw datasets do not provide clear insights about vehicle pricing, brand performance, fuel types, or manufacturing trends.

The purpose of this project is to convert raw data into an interactive dashboard that allows users to quickly understand important information and make informed decisions through visual analysis.

Objective

The objectives of this project are:

- To import and clean the Cars Dataset.
- To prepare the data for analysis.
- To create Pivot Tables for summarizing data.
- To develop KPI Cards for quick performance measurement.
- To create Pivot Charts for visual representation.
- To add slicers for interactive filtering.
- To design a professional Excel dashboard.
- To generate meaningful business insights.

Dataset Description

The dashboard was developed using the Cars Dataset downloaded from Kaggle.

<https://www.kaggle.com/datasets/aishwaryamuthukumar/cars-dataset-audi-bmw-ford-hyundai-skoda-vw>

The dataset contains information related to different car models and includes the following fields:

- Make
- Model
- Year
- Price
- Mileage
- Fuel Type
- Transmission
- Engine Size

The dataset is suitable for analyzing vehicle availability, pricing trends, fuel preferences, transmission types, and manufacturing year distribution.

Workflow

Step 1: Dataset Collection

- Download the Cars Dataset from Kaggle.

	A	B	C	D	E	F	G	H	I	J
1	model	year	price	transmission	mileage	fuelType	tax	mpg	engineSize	Make
2	A1	2017	12500	Manual	15735	Petrol	150	55.4	1.4	audi
3	A6	2016	16500	Automatic	36203	Diesel	20	64.2	2	audi
4	A1	2016	11000	Manual	29946	Petrol	30	55.4	1.4	audi
5	A4	2017	16800	Automatic	25952	Diesel	145	67.3	2	audi
6	A3	2019	17300	Manual	1998	Petrol	145	49.6	1	audi
7	A1	2016	13900	Automatic	32260	Petrol	30	58.9	1.4	audi
8	A6	2016	13250	Automatic	76788	Diesel	30	61.4	2	audi
9	A4	2016	11750	Manual	75185	Diesel	20	70.6	2	audi
10	A3	2015	10200	Manual	46112	Petrol	20	60.1	1.4	audi
11	A1	2016	12000	Manual	22451	Petrol	30	55.4	1.4	audi
12	A3	2017	16100	Manual	28955	Petrol	145	58.9	1.4	audi
13	A6	2016	16500	Automatic	52198	Diesel	125	57.6	2	audi
14	Q3	2016	17000	Manual	44915	Diesel	145	52.3	2	audi
15	A3	2017	16400	Manual	21695	Petrol	30	58.9	1.4	audi
16	A6	2015	15400	Manual	47348	Diesel	30	61.4	2	audi
17	A3	2017	14500	Automatic	26156	Petrol	145	58.9	1.4	audi
18	Q3	2016	15700	Automatic	28396	Diesel	145	53.3	2	audi
19	A3	2014	13900	Automatic	30516	Petrol	30	56.5	1.4	audi
20	Q5	2016	19000	Automatic	37652	Diesel	200	47.1	2	audi
21	Q3	2014	17000	Manual	34110	Petrol	145	47.9	1.4	audi
22	A5	2017	17300	Automatic	37100	Diesel	145	67.3	2	audi
23	Q5	2016	20200	Automatic	23714	Diesel	200	47.1	2	audi
24	A5	2017	22500	Automatic	21649	Diesel	145	58.9	3	audi
25	A6	2015	17500	Automatic	62748	Diesel	150	53.3	3	audi
26	Q5	2016	20000	Automatic	23789	Diesel	200	47.1	2	audi
27	A4	2017	18500	Automatic	17418	Diesel	145	62.8	2	audi
28	A5	2017	19500	Automatic	33300	Diesel	145	61.4	2	audi
29	A1	2018	15800	Manual	10793	Petrol	145	56.5	1.4	audi
30	A4	2018	17200	Automatic	25138	Diesel	145	70.6	2	audi
31	A6	2015	14000	Manual	61176	Diesel	30	62.8	2	audi
32	A1	2015	9500	Automatic	54754	Diesel	0	76.3	1.6	audi
33	Q3	2017	16500	Manual	21369	Petrol	125	51.4	1.4	audi
34	Q3	2014	15000	Automatic	39190	Diesel	205	47.9	2	audi

- Save it as an Excel Workbook (.xlsx).

Step 2: Data Import

- Open the dataset in Microsoft Excel.
- Convert the data into an Excel Table (Ctrl + T).

	A	B	C	D	E	F	G	H	I	J
1	model	year	price	transmission	mileage	fuelType	tax	mpg	engineSize	Make
2	A1	2017	12500	Manual	15735	Petrol	150	55.4	1.4	AUDI
3	A6	2016	16500	Automatic	36203	Diesel	20	64.2	2	AUDI
4	A1	2016	11000	Manual	29946	Petrol	30	55.4	1.4	AUDI
5	A4	2017	16800	Automatic	25952	Diesel	145	67.3	2	AUDI
6	A3	2019	17300	Manual	1998	Petrol	145	49.6	1	AUDI
7	A1	2016	13900	Automatic	32260	Petrol	30	58.9	1.4	AUDI
8	A6	2016	13250	Automatic	76788	Diesel	30	61.4	2	AUDI
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13	A6	2016	16500	Automatic	52198	Diesel	125	57.6	2	AUDI
14	Q3	2016	17000	Manual	44915	Diesel	145	52.3	2	AUDI
15	A3	2017	16400	Manual	21695	Petrol	30	58.9	1.4	AUDI
16	A6	2015	15400	Manual	47348	Diesel	30	61.4	2	AUDI
17	A3	2017	14500	Automatic	26156	Petrol	145	58.9	1.4	AUDI
18	Q3	2016	15700	Automatic	28396	Diesel	145	53.3	2	AUDI
19	A3	2014	13900	Automatic	30516	Petrol	30	56.5	1.4	AUDI
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21	Q3	2014	17000	Manual	34110	Petrol	145	47.9	1.4	AUDI
22	A5	2017	17300	Automatic	37100	Diesel	145	67.3	2	AUDI
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33	Q3	2017	16500	Manual	21369	Petrol	125	51.4	1.4	AUDI
34	Q3	2014	15000	Automatic	39190	Diesel	205	47.9	2	AUDI

Dataset imported and converted into an Excel Table

Step 3: Data Cleaning

The dataset was cleaned and prepared for analysis at this stage. See the *Data Cleaning Process* section for the full list of cleaning steps performed.

	A	B	C	D	E	F	G	H	I	J
1	model	year	price	transmission	mileage	fuelType	tax	mpg	engineSize	Make
2	A1	2017	12500	Manual	15735	Petrol	150	55.4	1.4	AUDI
3	A6	2016	16500	Automatic	36203	Diesel	20	64.2	2	AUDI
4	A1	2016	11000	Manual	29946	Petrol	30	55.4	1.4	AUDI
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7	A1	2016	13900	Automatic	32260	Petrol	30	58.9	1.4	AUDI
8	A6	2016	13250	Automatic	76788	Diesel	30	61.4	2	AUDI
9	A4	2016	11750	Manual	75185	Diesel	20	70.6	2	AUDI
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11	A1	2016	12000	Manual	22451	Petrol	30	55.4	1.4	AUDI
12	A3	2017	16100	Manual	28955	Petrol	145	58.9	1.4	AUDI
13	A6	2016	16500	Automatic	52198	Diesel	125	57.6	2	AUDI
14	Q3	2016	17000	Manual	44915	Diesel	145	52.3	2	AUDI
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16	A6	2015	15400	Manual	47348	Diesel	30	61.4	2	AUDI
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20	Q5	2016	19000	Automatic	37652	Diesel	200	47.1	2	AUDI
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29	A1	2018	15800	Manual	10793	Petrol	145	56.5	1.4	AUDI
30	A4	2018	17200	Automatic	25138	Diesel	145	70.6	2	AUDI
31	A6	2015	14000	Manual	61176	Diesel	30	62.8	2	AUDI
32	A1	2015	9500	Automatic	54754	Diesel	0	76.3	1.6	AUDI
33	Q3	2017	16500	Manual	21369	Petrol	125	51.4	1.4	AUDI
34	Q3	2014	15000	Automatic	39190	Diesel	205	47.9	2	AUDI

Dataset cleaned and prepared for analysis

Step 4: Create Pivot Tables

Create the following Pivot Tables:

- Cars by Brand
- Average Price by Brand
- Fuel Type Distribution
- Transmission Type
- Cars by Manufacturing Year
- Top 10 Expensive Models

	A	B	C	D	E	F	G	H
1	Cars by Brand			Average Price by Brand			Fuel Type Distribution	
2								
3	Row Labels	Count of model		Row Labels	Average of price		Row Labels	Count of model
4	AUDI	10564		AUDI	22853.95608		Diesel	28619
5	BMW	10664		BMW	22692.88869		Electric	5
6	FORD	17810		FORD	12270.10348		Hybrid	2887
7	HYUNDAI	4774		HYUNDAI	12727.80938		Other	238
8	SKODA	6188		SKODA	14284.80268		Petrol	39843
9	TOYOTA	6699		TOYOTA	12529.79907		Grand Total	71592
10	VOLKSWAGEN	14893		VOLKSWAGEN	16807.89807			
11	Grand Total	71592		Grand Total	16557.39563			
12								
13	Transmission Type			Cars by Year			Top 10 Expensive Models	
14								
15	Row Labels	Count of model		Row Labels	Count of model		Row Labels	Average of price
16	Automatic	13926		1996	2		8 Series	63997.79487
17	Manual	42471		1997	2		California	57593.86667
18	Other	4		1998	4		i8	57012.58824
19	Semi-Auto	15191		1999	5		M5	57760
20	Grand Total	71592		2000	6		Q8	60115.01449
21				2001	8		R8	97652.21429
22				2002	17		RS5	50703.60714
23				2003	22		RS6	55963.87179
24				2004	32		Supra	50741
25				2005	52		X7	69842.76364
26				2006	67		Grand Total	63259.24471
27				2007	114			
28				2008	156			
29				2009	222			
30				2010	234			
31				2011	322			
32				2012	480			
33				2013	1978			
34				2014	3031			
35				2015	5543			

Pivot Tables summarizing the dataset

Step 5: Create KPI Cards

The following KPI Cards were created:

- Total Cars
- Average Price
- Average Mileage
- Average Engine Size

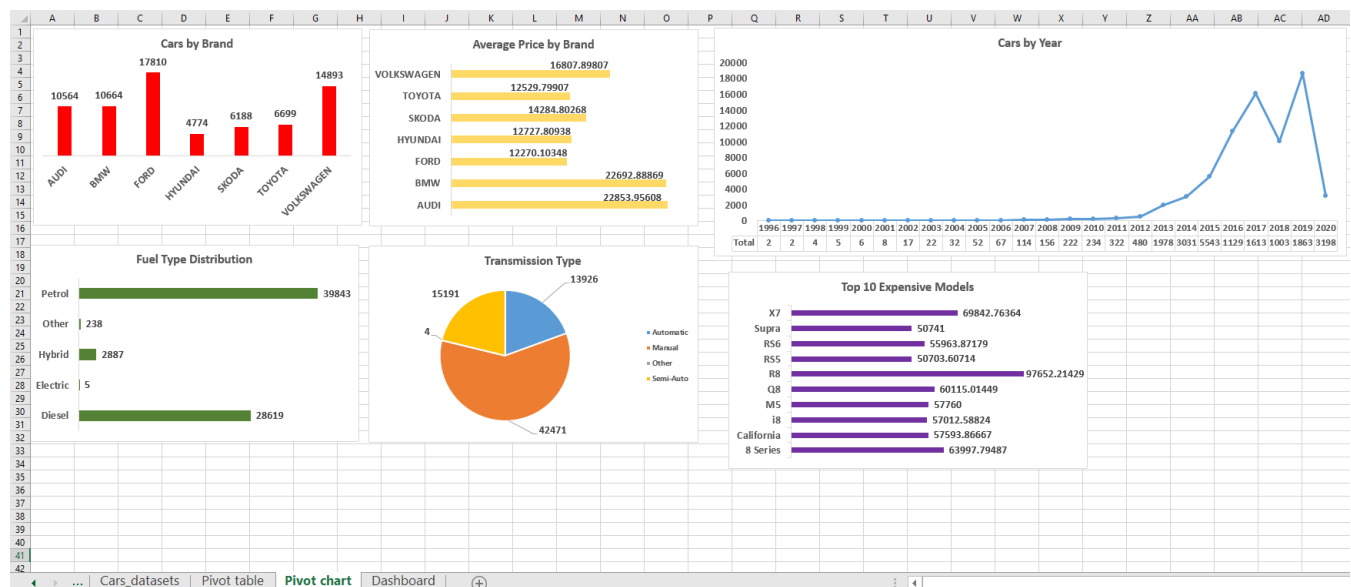
Count of model
71592
Average of price
16557.39563
Average of mileage
23270.37608
Average of engineSize
1.636570614

KPI Cards for the dashboard

Step 6: Create Pivot Charts

The following charts were created:

Chart	Chart Type
Cars by Brand	Column Chart
Average Price by Brand	Bar Chart
Fuel Type Distribution	Bar Chart
Transmission Type	Pie Chart
Cars by Year	Line Chart
Top 10 Expensive Models	Horizontal Bar Chart



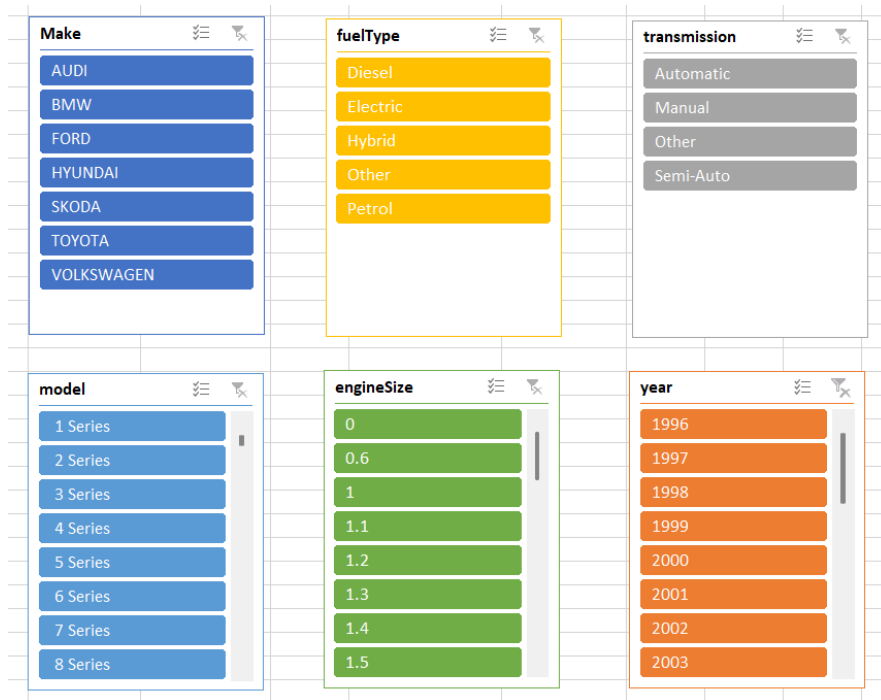
Pivot Charts built from the Pivot Tables

Step 7: Add Slicers

Insert slicers for:

- Make
- Model
- Fuel Type
- Transmission
- Engine Size
- Year

These slicers allow users to filter all Pivot Tables and Charts dynamically.

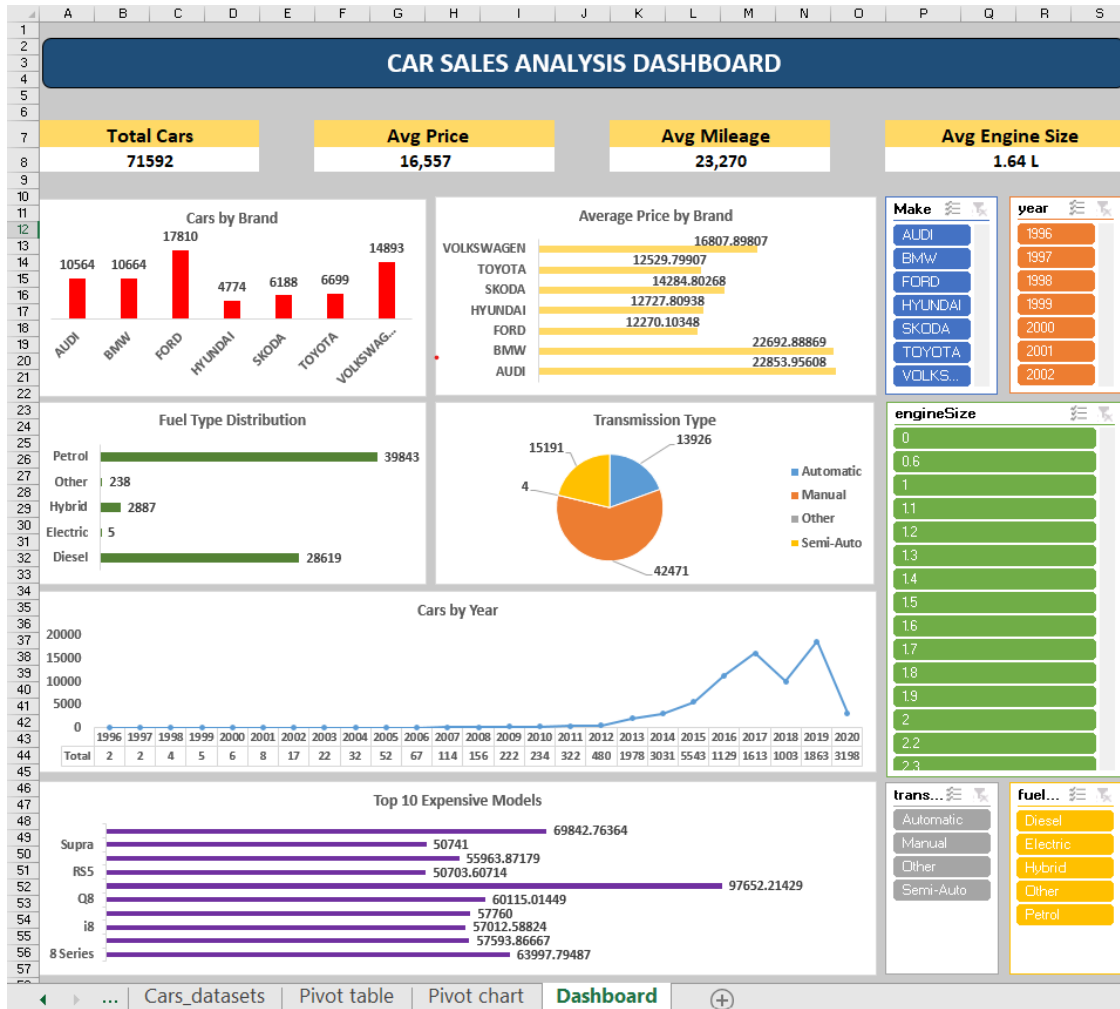


Slicers added for interactive filtering

Step 8: Dashboard Design

Arrange all dashboard elements professionally.

- KPI Cards at the top
- Charts in the center
- Slicers on the right or left side
- Apply consistent colors and formatting



Final Dashboard Layout

Step 9: Dashboard Analysis

Analyze the dashboard to identify trends and patterns.

The dashboard helps users to:

- Compare car brands.
- Analyze average prices.
- Study fuel type distribution.
- Compare transmission types.
- Observe yearly trends.
- Identify premium vehicle models.

Data Cleaning Process

The dataset was cleaned before analysis to improve data quality.

The following operations were performed:

- Imported the CSV dataset.
- Saved the file as an Excel Workbook.
- Checked for blank cells.
- Removed duplicate records.
- Converted the dataset into an Excel Table.
- Verified column data types.
- Organized the dataset for Pivot Table creation.

Dashboard KPIs

The dashboard includes four important KPI Cards.

Total Cars

Displays the total number of cars available in the dataset.

Average Price

Shows the average selling price of all vehicles.

Average Mileage

Displays the average mileage of vehicles.

Average Engine Size

Shows the average engine capacity.

These KPIs provide an instant overview of the dataset.

Charts and Visualizations

The dashboard contains multiple Pivot Charts for better visualization.

Cars by Brand

Shows the number of vehicles available for each manufacturer.

Average Price by Brand

Compares average selling prices among brands.

Fuel Type Distribution

Displays Petrol, Diesel, Hybrid, and other fuel types.

Transmission Type

Compares Manual and Automatic vehicles.

Cars by Manufacturing Year

Shows the yearly distribution of vehicles.

Top 10 Expensive Models

Displays the highest-priced vehicle models.

Slicers and Filters

Interactive slicers improve dashboard usability.

The following slicers were added:

- Make
- Fuel Type
- Transmission
- Engine Size
- Year

Users can instantly filter all charts and KPI Cards using these slicers.

Insights from the Dashboard

The dashboard provides several valuable insights.

- Some brands have significantly more vehicle listings than others.
- Average prices differ considerably among manufacturers.
- Petrol and Diesel vehicles dominate the dataset.
- Transmission preferences can be compared easily.
- Vehicle availability changes across manufacturing years.
- Premium models are identified through the Top 10 Expensive Models chart.
- Interactive slicers allow detailed and customized analysis.

Conclusion

The Car Sales Dashboard successfully transforms raw vehicle data into an interactive reporting system using Microsoft Excel.

By utilizing Pivot Tables, Pivot Charts, KPI Cards, and Slicers, the dashboard enables users to analyze data efficiently and make informed decisions. The project demonstrates practical Excel dashboard development skills and highlights the importance of data visualization in business intelligence and data analytics.